LESSON 7.4

Compare and Order Integers and Rational Numbers

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare and order integers using a number line.

WHY THIS MATTERS Lucy and Yuzuki are playing a trivia game with these rules: if a player answers a question correctly, they receive 1 point; if they answer incorrectly, they lose 0.25 point. The player with the greatest score wins the game. The final scores for two games are shown in the table. Who is the winner of each game?

REMEMBER

Compare and Order Integers and Rational Numbers

Comparing & Ordering Rational Numbers. Formula: Left is Less (<) | Right is Greater (>) To compare integers means to see which one is greater or less. On a number line, numbers get bigger as you go right and smaller as you go left.

1. Plot or visualize numbers on a horizontal number line.
2. Numbers further to the left are always less than numbers to the right.
3. For negatives: A negative number with a larger absolute value is smaller (-10 < -2)

Words: **Number line** ***(Recta numérica)*** · **Compare** ***(Comparar)*** · **Order** ***(Ordenar)*** · **Greater than** ***(Mayor que)*** · **Less than** ***(Menor que)***

1 **Circle the letter of the best answer. Show how you know.**

What is the absolute value of -9?

(A) -9
(B) 0
(C) 1
(D) 9

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

(A) -5
(B) -3
(C) 2
(D) 7

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is |0|?

(A) 1
(B) -1
(C) 0
(D) There is no answer

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A weather station recorded these temperatures during a winter week: Monday -4°F, Tuesday 2°F, Wednesday -7°F, Thursday -1°F, Friday 5°F. The station needs to report the coldest and warmest days. Which day was coldest?

(A) Thursday at −1°F
(B) Tuesday at 2°F
(C) Monday at −4°F
(D) Wednesday at −7°F

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A weather station recorded these temperatures during a winter week: Monday -4°F, Tuesday 2°F, Wednesday -7°F, Thursday -1°F, Friday 5°F. The station needs to report the coldest and warmest days. Which is colder, −4°F or −1°F?

(A) −4°F, because it is farther left on the number line

(B) −1°F, because 1 is smaller than 4

(C) They are the same

(D) −1°F, because it is closer to zero

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A weather station recorded these temperatures during a winter week: Monday -4°F, Tuesday 2°F, Wednesday -7°F, Thursday -1°F, Friday 5°F. The station needs to report the coldest and warmest days. Order the temperatures from coldest to warmest.

(A) −1, −4, −7, 2, 5

(B) 2, 5, −1, −4, −7

(C) 5, 2, −1, −4, −7

(D) −7, −4, −1, 2, 5

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does plotting on the number line help you compare and order integers?

SENTENCE STARTER On the number line, ___ is to the left of ___, so ___ < ___. The order from least to greatest is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can compare and order integers using a number line.

WHY THIS MATTERS Lucy and Yuzuki are playing a trivia game with these rules: if a player answers a question correctly, they receive 1 point; if they answer incorrectly, they lose 0.25 point. The player with the greatest score wins the game. The final scores for two games are shown in the table. Who is the winner of each game?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each treasure marker into the right zone on the map: below sea level or above sea level.

Groups: **Less than 0 (below sea level)** **Greater than 0 (above sea level)**

_____ -6 m shipwreck

_____ 4 m cliff nest

_____ -1 m anchor

_____ 7 m crow's nest

_____ -12 m trench chest

_____ 3 m palm flag

2 **Write the letter of the matching item on each line.**

Match each treasure-map comparison to the symbol that makes it true.

_____ -6 ___ -2

_____ 5 ___ -9

_____ -4 ___ -11

_____ -7 ___ 0

(A) <

(B) >

(C) >

(D) <

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Four spots: -3 m, 2 m, -7 m, 0 m. Order them from least to greatest.

(A) -7, -3, 0, 2

(B) -7, -3, 2, 0

(C) -3, -7, 0, 2

(D) 2, 0, -3, -7

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Three buried chests sit at -9 m, -4 m, and -15 m below sea level. Which chest is at the GREATEST elevation (closest to the surface)?

(A) -9 m

(B) -15 m

(C) They are equal

(D) -4 m

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

On the map, a sunken lantern is at -8 m and a coral shelf is at -3 m. Which symbol makes the comparison true: -8 ___ -3 ?

(A) >

(B) =

(C) ≥

(D) <

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — The Mis-Ranked Treasure Map

1. **Problem** Order the chest depths from least to greatest: -8 m and -15 m.
2. **Crew thinking** 15 has bigger digits than 8, so -15 must be the bigger number. I'll list -8 first because it must be smaller.
3. **Crew answer** Least to greatest: -8, -15

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does plotting on the number line help you compare and order integers?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help